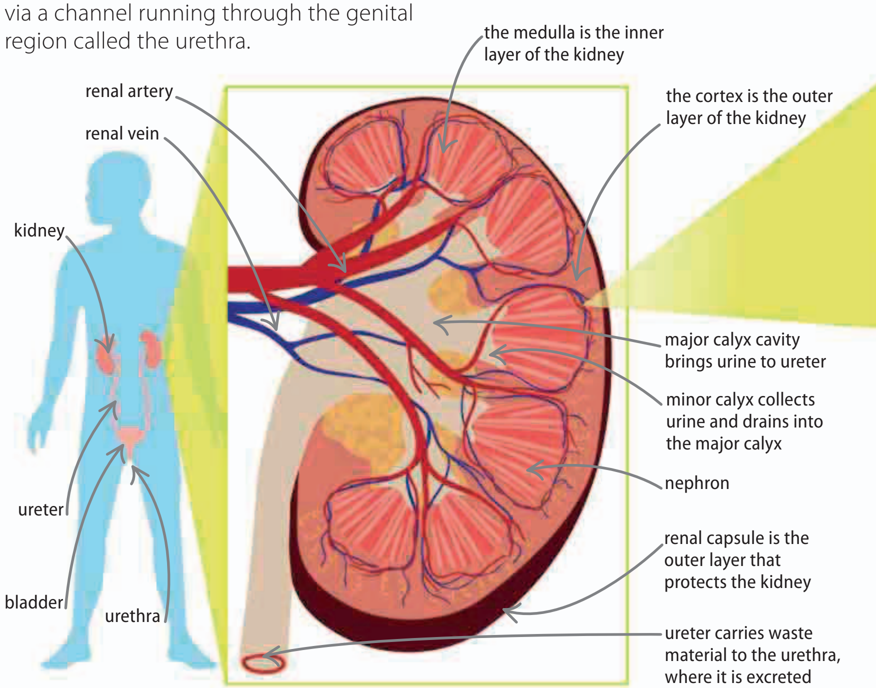
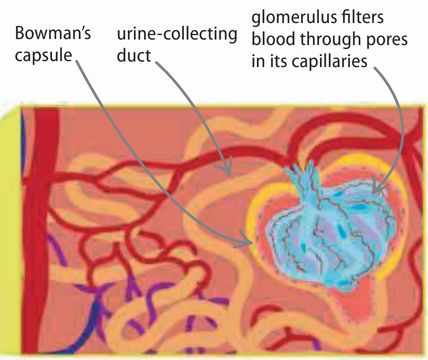
Kidneys and bladder

In humans—and other vertebrates—most waste products are filtered from the blood supply by the kidneys. The liquid produced—known as urine—trickles from each kidney through a long tube called a ureter. Both ureters empty into the bladder, a flexible bag in the pelvic region. When this is about half full, the weight of the liquid creates the urge to urinate. Urine is expelled from the bladder via a channel running through the genital region called the urethra.

▽ Inside the kidneys
A renal artery brings waste-filled blood to the kidney. The blood is dispersed to the outer regions, called the cortex, where the filtering happens in thousands of tiny units called nephrons. From there, the clean blood is returned to the body via a renal vein. Drops of the filtered waste are collected by the calyx, a multiheaded funnel that connects to the ureter.



Even **water** can be toxic, because too much in the body causes the brain to swell and can kill.



△ Nephron
Tiny blood vessels form into a netlike structure called a glomerulus. The liquid portion of the blood squirts out through the thin walls of the glomerulus into a bell-shaped Bowman's capsule. The solid blood cells cannot escape, but the waste material travels with the liquid through a series of tubules (tiny tubes) to a collecting duct that leads back through the medulla to the ureter.

Osmoregulation

The kidneys also carry out osmoregulation, controlling the amount of water in the body. When there is a lack of water, the nephron tubules reabsorb some of it from urine so it is not expelled unnecessarily. Osmoregulation is governed by a hormone called antidiuretic hormone, or ADH, which is produced by the pituitary gland.

▷ Rising and falling
The levels of ADH in the blood are constantly adjusting to maintain the right amount of water in the blood in a cycle, shown here.